



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Features

1/6-Inch 1.3Mp System-On-A-Chip (SOC) CMOS Digital Image Sensor

MT9M113

For the latest data sheet, refer to Micron's Web site: www.micron.com/imaging

Features

- DigitalClarity[®] CMOS imaging technology
- Superior low-light performance
- Ultra-low-power, low-cost
- Internal master clock generated by on-chip phase-locked loop (PLL) oscillator
- Electronic rolling shutter (ERS), progressive scan
- Integrated image flow processor (IFP) for single-die camera module
- Automatic image correction and enhancement, including four-channel lens shading correction
- Arbitrary image scaling with anti-aliasing
- Two-wire serial interface providing access to registers and microcontroller memory
- Selectable output data format: YCbCr, 565RGB, 555RGB, 444RGB, processed Bayer, RAW8- and RAW10-bit
- Output FIFO for data rate equalization
- Programmable I/O slew rate
- Parallel and serial mobile industry processor interface (MIPI) data output
- Xenon and LED flash support with fast exposure adaptation
- Independently configurable gamma correction

Applications

- Cellular phones
- PC cameras
- PDAs

Table 1: Key Performance Parameters

Parameter		Value
Optical format		1/6-inch (5:4)
Full resolution		1280 x 1024 pixels (SXGA)
Pixel size		1.75µm x 1.75µm
Dynamic range		66dB (preliminary)
SNR MAX		38.5dB (preliminary)
Responsivity		0.54 V/lux-sec (preliminary)
Chief ray angle (CRA)		24.63° MAX at 90% image height
Color filter array		RGB Bayer pattern
Active pixel array area		2.28mm x 1.83mm
Shutter type		Electronic rolling shutter (ERS)
Input clock frequency		8–54 MHz
Maximum frame rate		15 fps at full resolution, 24 fps in preview mode, 30 fps in video mode
Maximum pixel data output		30 Mp/s
Maximum pixel clock frequency		60 MHz
Supply voltage	Analog	2.5–3.1V
	Digital	1.7–1.95V
	I/O	1.7–3.1V
	PLL	2.5–3.1V
	MIPI	1.7–1.95V
ADC resolution		10-bit, on-die
Power consumption		222mW at 15 fps, full resolution
		156mW at 30 fps, video mode
		TBD at 24 fps, preview mode
		TBD, standby
Operating temperature		–30°C to +70°C (at junction)

Ordering Information

Table 2: Available Part Numbers

Part Number	Description
MT9M113D00STCK24AC1	Bare die



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Table of Contents

Table of Contents

Features.....	1
Applications.....	1
Ordering Information.....	1
Table of Contents.....	2
List of Figures.....	5
General Description.....	7
MT9M113 Overview.....	7
Signal Description.....	9
Typical Connections.....	10
Architecture Overview.....	11
Sensor Core Description.....	11
Pixel Array.....	12
Default Readout Order.....	13
Analog Processing.....	14
Analog Readout Channel.....	14
Gain Options.....	14
Integration Time.....	14
PLL.....	15
PLL Generated Master Clock.....	15
PLL Setup.....	15
Digital Processing.....	15
Readout Options.....	15
Window Size.....	15
Readout Modes.....	15
Horizontal Mirror.....	15
Vertical Flip.....	16
Column and Row Skip.....	16
Pixel Readouts.....	17
Programming Restrictions When Skipping.....	18
Binning.....	19
Binning Limitations.....	20
Raw Data Format.....	20
Raw Data Timing.....	21
SOC Description.....	22
Image Flow Processor.....	22
Test Patterns.....	24
First Black Level Subtraction and Digital Gain.....	25
Shading Correction (SC).....	25
The Correction Function.....	25
Defect Correction and Noise Reduction.....	25
Color Interpolation and Edge Detection.....	25
Color Correction and Aperture Correction.....	26
Image Cropping.....	26
Gamma Correction.....	26
Special Effects.....	27
RGB to YUV Conversion.....	27
Color Kill.....	27
YUV Color Filter.....	27
Image Scaling.....	27
YUV-to-RGB/YUV Conversion and Output Formatting.....	28
Color Conversion Formulas.....	28



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor

Table of Contents

Y'U'V'	28
Y'Cb'Cr' Using sRGB Formulas	28
Y'U'V' Using sRGB Formulas	28
Output Interface	29
Parallel and MIPI Output	29
Output Format and Timing	30
YUV/RGB Uncompressed Output	30
Uncompressed YUV/RGB Data Ordering	30
Uncompressed 10-Bit Bypass Output	31
FIFO	31
Camera Control	32
General Purpose I/Os	32
Output Enable Control	32
One-Time Programmable Memory	32
Firmware Architecture	34
Sequencer	34
Context and Operational Modes	34
Preview Mode	34
Still Capture and Video Modes	35
Snapshot and Flash	35
Video	35
Auto Exposure	35
AE Driver	36
Evaluative Algorithm	36
Accelerated Settling During Overexposure	36
Exposure Control	36
Auto White Balance	37
Flicker Detection	37
Two-Wire Serial Interface	38
Protocol	38
Start Condition	38
Stop Condition	38
Data Transfer	38
Slave Address/Data Direction Byte	38
Message Byte	39
Acknowledge Bit	39
No-Acknowledge Bit	39
Typical Serial Transfer	39
Single READ from Random Location	39
Single READ from Current Location	40
Sequential READ, Start from Random Location	40
Sequential READ, Start from Current Location	41
Single WRITE to Random Location	41
Sequential WRITE, Start at Random Location	42
Registers and Variables	43
How to Access Registers and Variables	43
Registers	43
Variables	43
Special Function Registers and MCU SRAM	44
Core Registers	45
Double-buffered Registers	45
Bad Frames	45
Changes to Integration Time	45



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor

Table of Contents

Changes to Gain Settings	45
Register Summary	47
Register Description	58
Timing Specifications	103
Power-up sequence	103
Power on Reset	104
Reset	105
Hard Reset	105
Soft Reset	106
Standby Modes	107
Hard Standby	107
Soft Standby	108
Signal States	109
Spectral Characteristics	110
Electrical Specifications	111
Two-Wire Serial Bus Timing	118
Revision History	121



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor List of Figures

List of Figures

Figure 1:	Typical Configuration1 (connection)	10
Figure 2:	SOC Block Diagram	11
Figure 3:	Sensor Core Block Diagram	12
Figure 4:	Pixel Color Pattern Detail (Top Right Corner)	13
Figure 5:	Imaging a Scene	13
Figure 6:	6 Pixels in Normal and Column Mirror Readout Modes	15
Figure 7:	6 Rows in Normal and Row Mirror Readout Modes	16
Figure 8:	8 Pixels in Normal and Column Skip 2X Readout Modes.	16
Figure 9:	Pixel Readout (no skipping)	17
Figure 10:	Pixel Readout (x_odd_inc = 3, y_odd_inc = 1)	17
Figure 11:	Pixel Readout (x_odd_inc = 1, y_odd_inc = 3)	18
Figure 12:	Pixel Readout (x_odd_inc = 3, y_odd_inc = 3)	18
Figure 13:	Pixel Readout (x_odd_inc = 3, y_odd_inc = 1, x_bin = 1)	19
Figure 14:	Pixel Readout (x_odd_inc = 3, y_odd_inc = 3, x_ybin = 1)	20
Figure 15:	Valid Image Data.	21
Figure 16:	Pixel Data Timing Example.	21
Figure 17:	Image Flow Processor	23
Figure 18:	Color Bar Test Pattern	24
Figure 19:	Gamma Correction Curve.	27
Figure 20:	Timing of Uncompressed Full Frame Data or Scaled Data Passing Through the FIFO	30
Figure 21:	Sequence of Signals for OTP Memory Operation	33
Figure 22:	Software Architecture	34
Figure 23:	Single READ from Random Location	40
Figure 24:	Single Read from Current Location	40
Figure 25:	Sequential READ, Start from Random Location	40
Figure 26:	Sequential READ, Start from Current Location	41
Figure 27:	Single WRITE to Random Location.	41
Figure 28:	Sequential WRITE, Start at Random Location	42
Figure 29:	Power Up Sequence	103
Figure 30:	Power On Reset Sequence	104
Figure 31:	Hard Reset Signal Sequence	105
Figure 32:	Soft Reset Signal Sequence	106
Figure 33:	Hard Standby Signal Sequence	107
Figure 34:	Soft Standby Signal Sequence	108
Figure 35:	Chief Ray Angle (CRA) vs. Image Height	110
Figure 36:	I/O Timing Diagram.	111
Figure 37:	Two-Wire Serial Bus Timing Parameters.	118



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor List of Tables

List of Tables

Table 1:	Key Performance Parameters.....	1
Table 2:	Available Part Numbers.....	1
Table 3:	Signal Descriptions.....	9
Table 4:	Row Address Sequencing (Sampling).....	19
Table 5:	Row Address Sequencing (Binning).....	20
Table 6:	Data Formats Supported by MIPI Interface.....	29
Table 7:	YCrCb Output Data Ordering.....	30
Table 8:	RGB Ordering in Default Mode.....	30
Table 9:	2-Byte RGB Format.....	31
Table 10:	Sensor Core Registers.....	47
Table 11:	SOC1 Registers.....	47
Table 12:	SOC2 Registers.....	48
Table 13:	SYSCTL Registers.....	51
Table 14:	XDMA Registers.....	51
Table 15:	Driver ID = 1: Sequencer Variables.....	51
Table 16:	Driver ID = 2: Auto Exposure Variables.....	52
Table 17:	Driver ID = 3: Auto White Balance Variables.....	53
Table 18:	Driver ID = 4: Anti-Flicker Variables.....	54
Table 19:	Driver ID = 7: Mode Variables.....	55
Table 20:	Driver ID = 11: Histogram Variables.....	56
Table 21:	GPIO Registers.....	57
Table 22:	Sensor Core Registers.....	58
Table 23:	SOC1 Registers.....	60
Table 24:	SOC2 Registers.....	64
Table 25:	SYSCTL Registers.....	72
Table 26:	XDMA Registers.....	76
Table 27:	Driver ID = 1: Sequencer Variables.....	78
Table 28:	Driver ID = 2: Auto Exposure Variables.....	84
Table 29:	Driver ID = 3: Auto White Balance Variables.....	86
Table 30:	Driver ID = 4: Anti-Flicker Variables.....	90
Table 31:	Driver ID = 7: Mode Variables.....	92
Table 32:	Driver ID = 11: Histogram Variables.....	99
Table 33:	GPIO Registers.....	101
Table 34:	Power-Up Signal Timing.....	103
Table 35:	POR Parameters.....	104
Table 36:	Hard Reset Signal Timing.....	105
Table 37:	Soft Reset Signal Timing.....	106
Table 38:	Hard Standby Signal Timing.....	107
Table 39:	Soft Standby Signal Timing.....	108
Table 40:	Status of Signals During Different States.....	109
Table 41:	I/O Timing Specifications.....	111
Table 42:	DC Electrical Definitions and Characteristics—Parallel Mode.....	113
Table 43:	DC Electrical Definitions and Characteristics—Serial Mode.....	114
Table 44:	I/O Parameters.....	116
Table 45:	Power Consumptions at Different Resolutions.....	116
Table 46:	Absolute Maximum Ratings.....	117
Table 47:	Two-Wire Serial Bus Characteristics.....	119



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor General Description

General Description

The Micron® Imaging MT9M113 is a 1/6-inch 1.3Mp CMOS digital image sensor with an integrated advanced camera system. This camera system features a microcontroller (MCU), a sophisticated IFP, and both parallel and serial MIPI interfaces. It also includes a one-time programmable memory for storing module specific information for identification purposes. The microcontroller manages all components of the camera system and sets key operation parameters for the sensor core to optimize the quality of raw image data entering the IFP. The sensor core consists of an active pixel array of 1300 x 1044 pixels, programmable timing and control circuitry including a PLL and external flash support, analog signal chain with automatic offset correction and programmable gain, and a 10-bit analog-to-digital converter (ADC). The entire SOC has ultra-low power requirements and superior low light performance that is particularly suitable for mobile applications. The MT9M113 is based on DigitalClarity—Micron's breakthrough low-noise CMOS imaging technology that achieves CCD-like image quality (based on signal-to-noise ratio and low-light sensitivity) while maintaining the inherent size, cost, power consumption, and integration advantages of CMOS.

MT9M113 Overview

The MT9M113 has a color image sensor with a Bayer color filter arrangement and a 1.3Mp active-pixel array with electronic rolling shutter (ERS). The sensor core readout is 10-bit, supports skipping and binning, and can be flipped and/or mirrored. The sensor core also supports separate analog and digital gain for all four color channels (R, Gr, Gb, B).

The MT9M113 also has an embedded PLL that can generate the internal sensor clock from the common wireless system clock. When in use, the PLL adjusts the incoming clock frequency up, allowing the MT9M113 to run at almost any desired resolution and frame rate within the sensor's capabilities. The PLL can be bypassed and powered down to reduce power consumption.

Low power consumption is a very important requirement for all components of wireless devices. The MT9M113 has numerous power-conserving features, including an internal soft standby mode and a hard standby mode, which allows the internal power bus to be disabled.

Another important consideration for wireless devices is their electromagnetic interference (EMI). The MT9M113 can be used with either a serial MIPI interface or the parallel data output interface, which has a programmable I/O slew rate to minimize EMI and an output FIFO to eliminate output data bursts.

The IFP and flexible programmability of the MT9M113 provide a variety of ways to enhance and optimize the image sensor's performance. Built-in optimization algorithms enable the MT9M113 to operate at factory settings as a fully automatic, highly adaptable camera; however, most of its settings are user-programmable.

These algorithms include black level conditioning, shading correction, defect correction, noise reduction, color interpolation, edge detection, color correction, aperture correction, and image formatting such as cropping and scaling.

The MT9M113 also includes a sequencer that coordinates all events triggered by the user. The sequencer manages auto white balance (AWB), flicker detection, and auto exposure (AE) for the different operating modes, which include preview, still capture, video, and snapshot with flash.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor MT9M113 Overview

The MT9M113 can operate in several modes including preview, still capture (snapshot), and video. All modes of operation are individually configurable and are organized as two contexts: context A and context B. A context is defined by sensor image size, frame rate, resolution, and other associated parameters. The user can switch between the two contexts by sending a command through the two-wire serial interface.

A two-wire serial register interface bus enables read/write access to control registers, variables, and special function registers within the MT9M113. The hardware registers include sensor core controls, color pipeline controls, and output controls. Variables are located in the microcontroller's RAM memory and are used for drivers such as those for AWB and AE. Special function registers are registers connected to the local bus of the microcontroller and include general purpose I/O (GPIO).

The GPIOs can be configured to allow the user to output a flash or achieve a 10-bit parallel output.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Signal Description

Signal Description

Table 3 provides the signal types and descriptions for the MT9M113. Power supply domains are also identified.

Table 3: Signal Descriptions

Name	Type	Description	Notes
STANDBY	Input	Controls sensor's standby mode, active HIGH.	
SCLK	Input	Two-wire serial interface clock.	
SADDR	Input	Selects device address for the two-wire serial interface. The address is 0x78 when SADDR is tied LOW, 0x7A if tied HIGH.	
RESET_BAR	Input	Master reset signal, active LOW (can leave floating if not used).	
EXTCLK	Input	Input clock signal 8–54 MHz.	
GPIO[4:0]	I/O	General purpose digital I/O, can be configured for FLASH/SHUTTER/DOUT_LSB0/DOUT_LSB1/MODULE_ID/OE_BAR.	
SDATA	I/O	Two-wire serial interface data.	
VPP	Input	High voltage programming pin for anti-fuse of module ID (must leave floating for normal operation).	
DOUT[7:0]	Output	Eight-bit image data output or most significant bits (MSB) of 10-bit sensor bypass mode.	
DOUT_N	Output	Differential MIPI data (sub-LVDS, negative) (must leave floating if not used).	
DOUT_P	Output	Differential MIPI data (sub-LVDS, positive) (must leave floating if not used).	
CLK_N	Output	Differential MIPI clock (sub-LVDS, negative) (must leave floating if not used).	
CLK_P	Output	Differential MIPI clock (sub-LVDS, positive) (must leave floating if not used).	
PIXCLK	Output	Pixel clock. Used for sampling DOUT, FRAME_VALID, and LINE_VALID.	
LINE_VALID	Output	Identifies pixels in the active line.	
FRAME_VALID	Output	Identifies rows in the active image.	
VDD	Supply	Digital power (1.8V typical).	
VAA_PIX	Supply	Pixel array power (2.8V typical).	
VAA	Supply	Analog power (2.8V typical).	
VDD_PLL	Supply	PLL power (2.8V typical).	
VDD_IO	Supply	I/O power supply (1.8V or 2.8V typical).	
GND_IO	Supply	I/O ground.	
DGND	Supply	Digital ground.	1
GND_PLL	Supply	PLL ground.	
AGND	Supply	Analog ground.	1
VDDIO_TX	Supply	I/O power supply for the MIPI interface 1.8V typical (must leave floating if not used).	

Notes: 1. AGND and DGND are not connected internally (inside the chip).

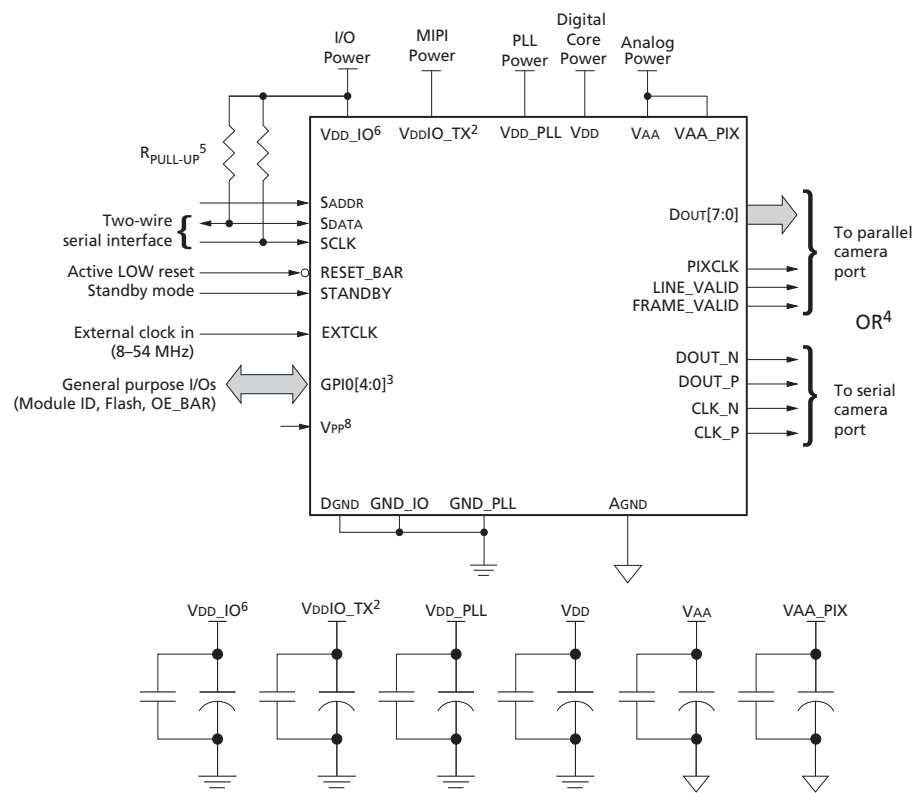


MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Typical Connections

Typical Connections

Figure 1 shows typical MT9M113 device connections. For low-noise operation, the MT9M113 requires separate power supplies for analog and digital circuitry. Incoming digital and analog ground conductors can be tied together next to the die. Both power supply rails should be decoupled from ground using capacitors as close as possible to the die. The use of inductance filters is not recommended on the power supplies or output signals. The MT9M113 also supports different digital core (VDD/DGND), MIPI (VDDIO_TX), and I/O (VDD_IO/GND_IO) power domains that can be at different voltages. The PLL requires a clean power source (VDD_PLL/GND_PLL).

Figure 1: Typical Configuration¹ (connection)



- Notes:
1. This typical configuration shows only one scenario out of multiple possible variations for this sensor.
 2. If a MIPI device is not required, the following pads must be left floating: DOUT_P, DOUT_N, CLK_P, CLK_N, and VDDIO_TX.
 3. The GPIO pads can serve multiple features that can be reconfigured. The function and direction will vary by applications.
 4. Only one of the output modes (serial or parallel) can be used at any time.
 5. A resistor value of 1.5KΩ is recommended for the two-wire serial interface R_{PULL-UP}⁵; however, greater values may be used for slower transmission speed.
 6. All inputs must be configured with VDD_IO.
 7. VAA and VAA_PIX must be tied together.
 8. VPP is the one-time programmable (OTP) memory signal and should be left floating during normal operation.

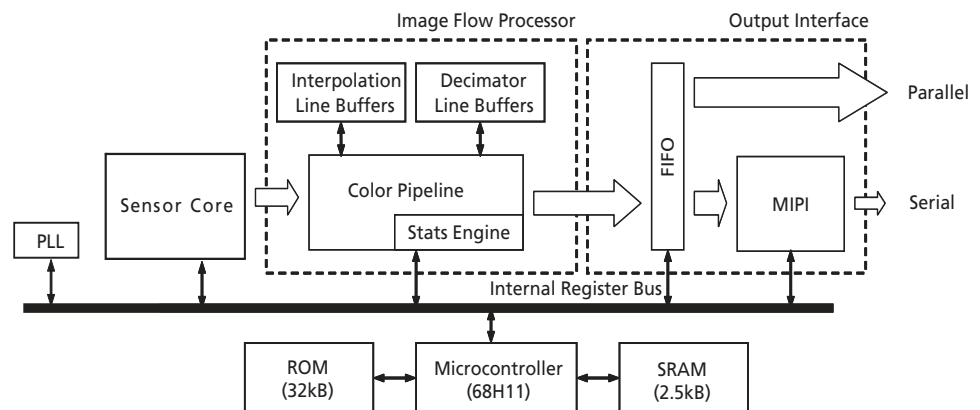


MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Architecture Overview

Architecture Overview

The MT9M113 combines a 1.3Mp sensor core together with an IFP to form a stand-alone solution that includes both image acquisition and processing. Both the sensor core and the IFP have internal registers that can be controlled by the user. In normal operation though, an integrated microcontroller controls most aspects of operation autonomously. The processed image data is transmitted to the host system either through a parallel bus or a serial data interface through the output interface.

Figure 2: SOC Block Diagram



Sensor Core Description

The sensor core of the MT9M113 is a progressive-scan sensor that generates a stream of pixel data at a constant frame rate, qualified by LINE_VALID (LV) and FRAME_VALID (FV). The maximum pixel rate is 30 Mp/s, corresponding to a pixel clock rate of 60 MHz. Figure 3 on page 12 shows a block diagram of the sensor core. It includes a 1.3-Mp active-pixel array. The timing and control circuitry sequences through the rows of the array, resetting and then reading each row in turn. In the time interval between resetting a row and reading that row, the pixels in the row integrate incident light. The exposure is controlled by varying the time interval between reset and readout. Once a row has been read, the data from the columns is sequenced through an analog signal chain (providing offset correction and gain), and then through an ADC. The output from the ADC is a 10-bit value for each pixel in the array.

The pixel array contains optically active and light-shielded (dark) pixels. The dark pixels are used to provide data for the offset-correction algorithms (black level control).

The sensor core contains a set of control and status registers that can be used to control many aspects of the sensor behavior including the frame size, exposure, and gain settings. These registers are controlled by the SOC firmware and can be accessed through a two-wire serial interface. Register values written to the sensor core may be overwritten by firmware.

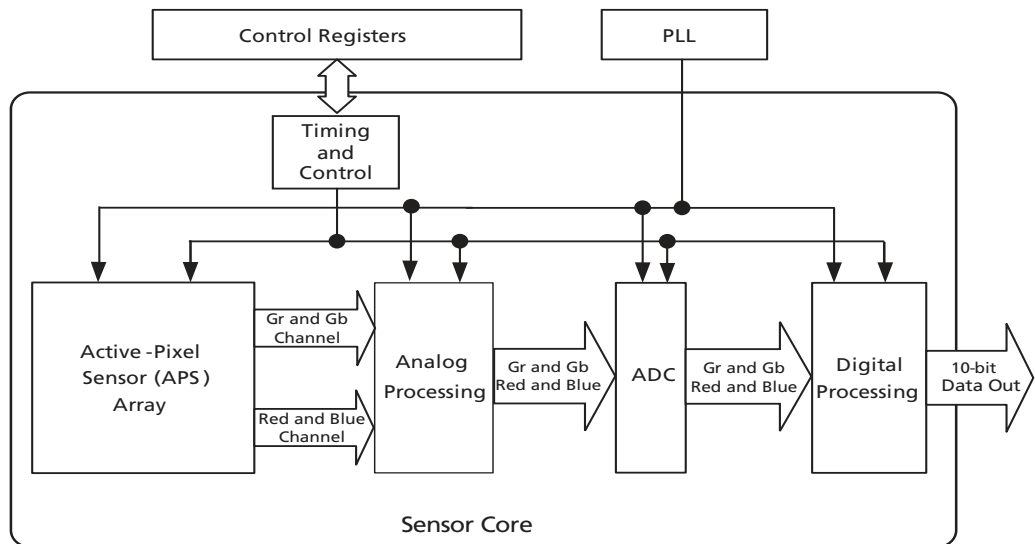
The output from the core is a Bayer pattern; alternate rows are a sequence of either green and red pixels or blue and green pixels. The offset and gain stages of the analog signal chain provide per-color control of the pixel data.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

A flash strobe output signal is provided to allow an external Xenon or LED light source to synchronize with the sensor exposure time. Additional I/O signals support the provision of an external mechanical shutter.

Figure 3: Sensor Core Block Diagram



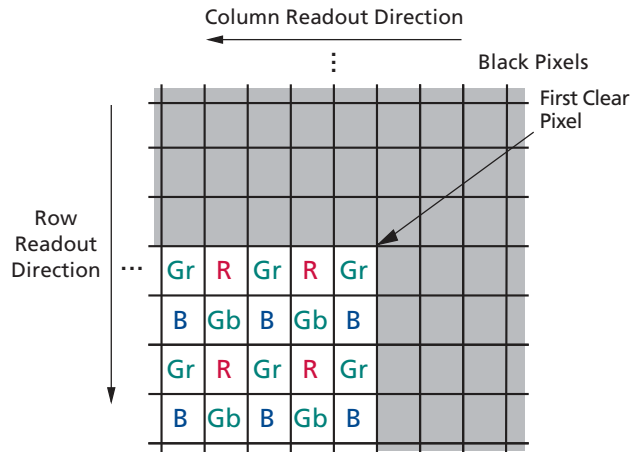
Pixel Array

The sensor core uses a Bayer color pattern, as shown in Figure 4 on page 13. The even-numbered rows contain green and red pixels; odd-numbered rows contain blue and green pixels. Even-numbered columns contain green and blue pixels; odd-numbered columns contain red and green pixels.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Figure 4: Pixel Color Pattern Detail (Top Right Corner)

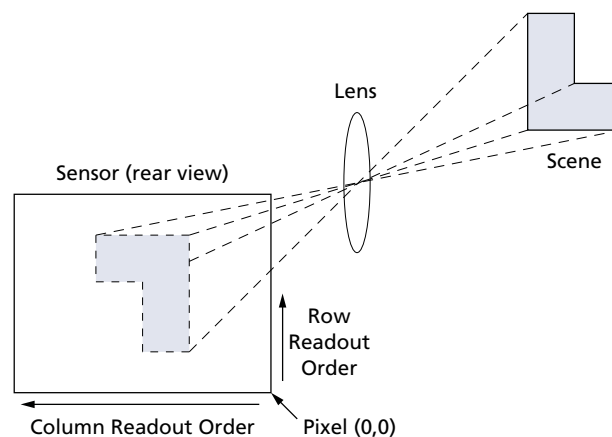


Default Readout Order

By convention, the sensor core pixel array is shown with pixel (0,0) in the top right corner. This reflects the actual layout of the array on the die. When the sensor is operating in a system, the active surface of the sensor faces the scene as shown in Figure 5.

When the image is read out of the sensor, it is read one row at a time, with the rows and columns sequenced. By convention, data from the sensor is shown with the first pixel read out in the case of the sensor core in the top left corner.

Figure 5: Imaging a Scene





MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Analog Processing

Analog Readout Channel

The sensor core features an analog readout channel as shown in Figure 3 on page 12. The readout channel consists of a gain stage, a sample-and-hold stage with black level calibration capability, and a 10-bit ADC.

Gain Options

The MT9M113 provides per-color gain control as well as the option of global gain control. Per-color and global gain control can be used interchangeably. A write to a global gain register is aliased as a write of the same data to the four associated color-dependent gain registers.

Integer digital gains in the range 0–7 can be programmed. A digital gain of 0 sets all pixel values to 0 (the pixel data will simply represent the value applied by the pedestal block). Gain settings are updated in every frame by the MCU auto feature; to make manual adjustments to gain settings, the MCU auto features must be disabled.

Integration Time

The integration time (exposure) of the MT9M113 is controlled by variables. While coarse integration time controls the integration duration steps of lines, fine integration time allows for sub-line accuracy. Integration time is updated in every frame by the MCU auto feature; to make manual adjustments to integration time, the MCU auto features must be disabled.

It is fundamental to the operation of an ERS that it is not possible to set an integration time that is greater than the frame time. Unlike earlier Micron Imaging parts, setting an integration time that is greater than the frame time does not affect the frame time; the behavior is undefined. On the MT9M113, it is not necessary to reprogram the frame time in order to make longer integration times available, as the frame time adjusts automatically. Long integration times increase the likelihood of image degradation due to increased accumulation of dark current.

If the integration time is changed while FV is asserted for frame n , the first frame output using the new integration time is frame $(n + 2)$. The sequence is as follows:

1. During frame n , the new integration time is held in the pending register.
2. At the start of frame $(n + 1)$, the new integration time is transferred to the live register. Integration for each row of frame $(n + 1)$ has been completed using the old integration time.
3. The earliest time that a row can start integrating using the new integration time is immediately after that row has been read for frame $(n + 1)$. The actual time that rows start integrating using the new integration time is dependent upon the new value of the integration time.
4. When frame $(n + 1)$ is read out, it will have been integrated using the new integration time.

If the integration time is changed on successive frames, each value written will be applied for a single frame; the latency between writing a value and it affecting the frame readout remains at two frames.

When the integration time and the gain are changed at the same time, the gain update is held off by one frame so that the first frame output with the new integration time also has the new gain applied.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

PLL

PLL Generated Master Clock

The PLL can generate a master clock signal whose frequency is up to 60 MHz (input clock from 8 through 54 MHz).

PLL Setup

Because the input clock frequency is unknown, the sensor starts up with the PLL disabled. The PLL takes time to power up. During this time, the behavior of its output clock signal is not guaranteed. The PLL output frequency is determined by two constants, M and N, and the input clock frequency.

Digital Processing

Readout Options

The sensor core supports different readout options to modify the image before it is sent to the IFP. The readout can be limited to a specific window of the original pixel array.

For preview modes, the sensor core supports both skipping and pixel averaging in x and y directions.

By changing the readout direction the image can be flipped in the vertical direction and/or mirrored in the horizontal direction.

Window Size

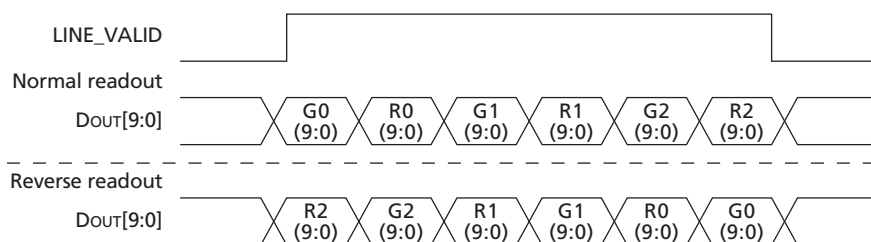
The image output size is set with registers `x_addr_start`, `x_addr_end`, `y_addr_start`, and `y_addr_end`. The edge pixels in the 1300 x 1044 array are present to avoid edge defects and should not be included in the visible window. Binning will change the image output size.

Readout Modes

Horizontal Mirror

When the sensor is configured to mirror the image horizontally, the order of pixel readout within a row is reversed, so that readout starts from `x_addr_end` and ends at `x_addr_start`. Figure 6 shows a sequence of 6 pixels being read out with normal readout and reverse readout. The SOC corrects for this change in sensor core output.

Figure 6: 6 Pixels in Normal and Column Mirror Readout Modes



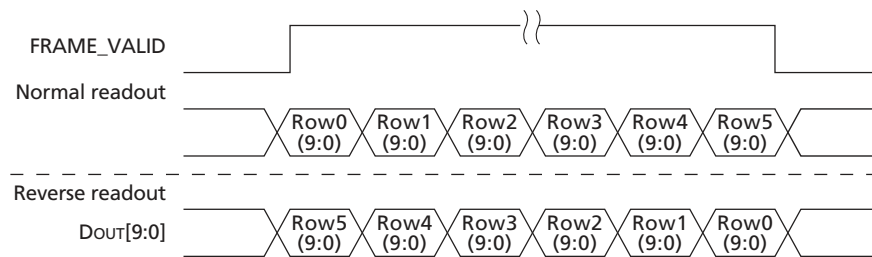


MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Vertical Flip

When the sensor is configured to flip the image vertically, the order in which pixel rows are read out is reversed, so that row readout starts from `y_addr_end` and ends at `y_addr_start`. Figure 7 shows a sequence of 6 rows being read out with normal readout and reverse readout. The SOC corrects for this change in sensor core output.

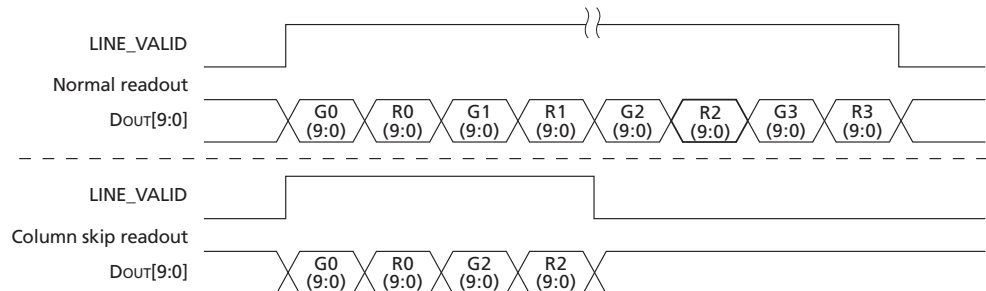
Figure 7: 6 Rows in Normal and Row Mirror Readout Modes



Column and Row Skip

The sensor core supports subsampling. Subsampling reduces the amount of data processed by the analog signal chain in the sensor and thereby allows the frame rate to be increased. This reduces the amount of row and column data processed and is equivalent to the skip2 readout mode provided by earlier Micron image sensors. Set the proper image output and crop sizes before enabling subsampling.

Figure 8: 8 Pixels in Normal and Column Skip 2X Readout Modes





MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Pixel Readouts

Figure 10 through Figure 12 on page 18 show a sequence of data being read out with no skipping, with $x_odd_inc = 3$ and $y_odd_inc = 1$, with $x_odd_inc = 1$ and $y_odd_inc = 3$, and with $x_odd_inc = 3$ and $y_odd_inc = 3$.

Figure 9: Pixel Readout (no skipping)

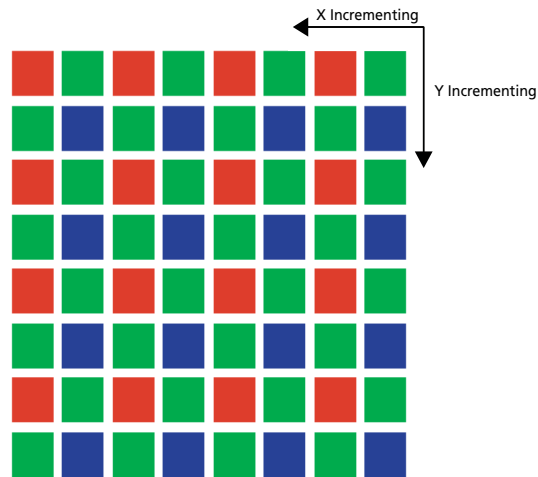
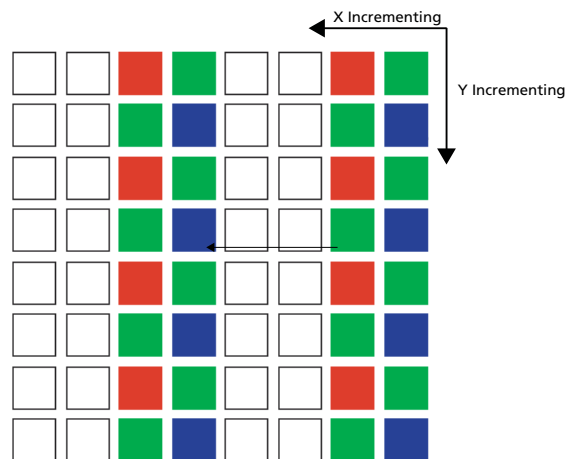


Figure 10: Pixel Readout ($x_odd_inc = 3$, $y_odd_inc = 1$)





MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Figure 11: Pixel Readout (x_odd_inc = 1, y_odd_inc = 3)

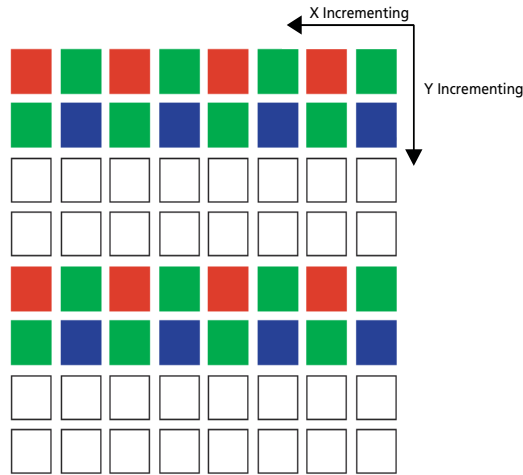
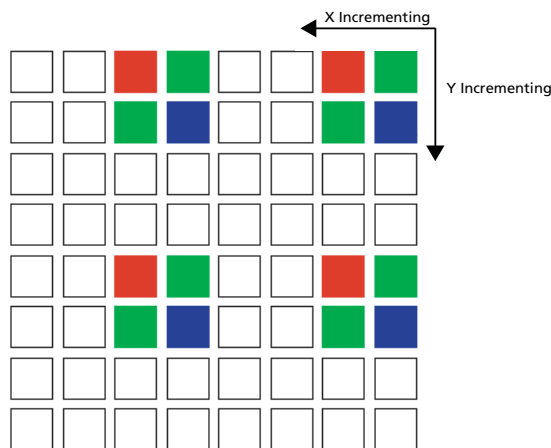


Figure 12: Pixel Readout (x_odd_inc = 3, y_odd_inc = 3)



Programming Restrictions When Skipping

When skipping is enabled as a viewfinder mode, and the sensor is switched back and forth between full resolution and skipping. It is recommended that `line_length_pck` be kept constant between the two modes. This allows the same integration times to be used in each mode.

When subsampling is enabled, it may be necessary to adjust the `x_addr_end` and `y_addr_end` settings. The values for these registers are required to correspond with rows/ columns that form part of the subsampling sequence. The adjustment should be made using the following rules:

$$\text{remainder} = (\text{addr_end} - \text{addr_start} + 1) \text{ AND } 4 \quad (\text{EQ } 1)$$

$$\text{if } (\text{remainder} == 0) \text{ addr_end} = \text{addr_end} - 2 \quad (\text{EQ } 2)$$



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Table 4 shows the row address sequencing for normal and subsampled (with $y_odd_inc = 3$) readout. The same sequencing applies to column addresses for subsampled readout. There are two possible subsampling sequences (because the subsampling sequence only reads half of the rows and columns) depending upon the alignment of the start address.

Table 4: Row Address Sequencing (Sampling)

Normal	Subsampled Sequence 1	Subsampled Sequence 2
0	0	No data
1	1	No data
2	No data	2
3	No data	3
4	4	No data
5	5	No data
6	No data	6
7	No data	7

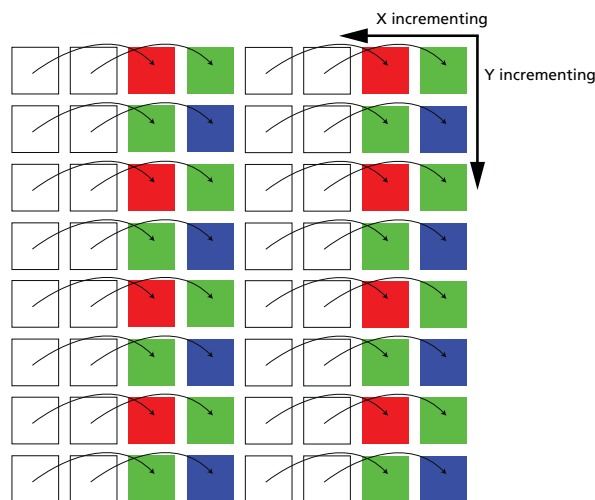
Binning

The MT9M113 sensor core supports 2 x 1 and 2 x 2 analog binning (column binning, also called x-binning and row/column binning, also called xy-binning). Binning has many of the same characteristics as subsampling but because it gathers image data from all pixels in the active window (rather than a subset of them), it achieves superior image quality and avoids the aliasing artifacts that can be a characteristic side effect of subsampling.

Binning is enabled by selecting the appropriate subsampling settings ($x_odd_inc = 3$ and $y_odd_inc = 1$ for x-binning, $x_odd_inc = 3$ and $y_odd_inc = 3$ for xy-binning) and setting the appropriate binning bit in read_mode register. As for subsampling, x_addr_end and y_addr_end may require adjustment when binning is enabled.

The effect of the different subsampling settings is shown in Figure 13 and Figure 14 on page 20.

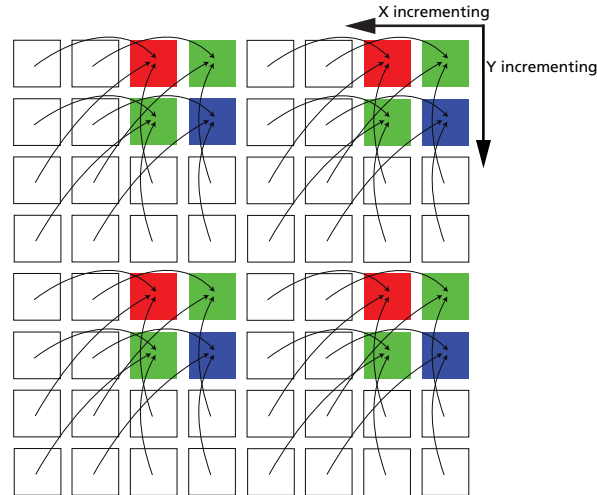
Figure 13: Pixel Readout ($x_odd_inc = 3$, $y_odd_inc = 1$, $x_bin = 1$)





MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Sensor Core Description

Figure 14: Pixel Readout ($x_odd_inc = 3$, $y_odd_inc = 3$, $x_ybin = 1$)



Binning Limitations

Binning requires a different sequencing of the pixel array and imposes different timing limits on the operation of the sensor. In particular, xy-binning requires two read operations from the pixel array for each line of output data, which has the effect of increasing the minimum line blanking time.

As a result, when xy-binning is enabled, some of the programming limits declared in the parameter limit registers are no longer valid. In addition, the default values for some of the manufacturer-specific registers need to be reprogrammed. None of these adjustments are required for x-binning. The sensor must be taken out of streaming mode before switching between binned and non-binned operation. The row addresses for various binning modes is shown in Table 5.

Table 5: Row Address Sequencing (Binning)

Normal	Binning Sequence 1	Binning Sequence 2
0	0,2	No data
1	1,3	No data
2	No data	2,4
3	No data	3,5
4	4,6	No data
5	5,7	No data
6	No data	6,8
7	No data	7,9

Raw Data Format

The sensor core image data is read out in a progressive scan. Valid image data is surrounded by horizontal blanking and vertical blanking, as shown in Figure 15 on page 21. The amount of horizontal blanking and vertical blanking is programmable. LV is HIGH during the shaded region of the figure.